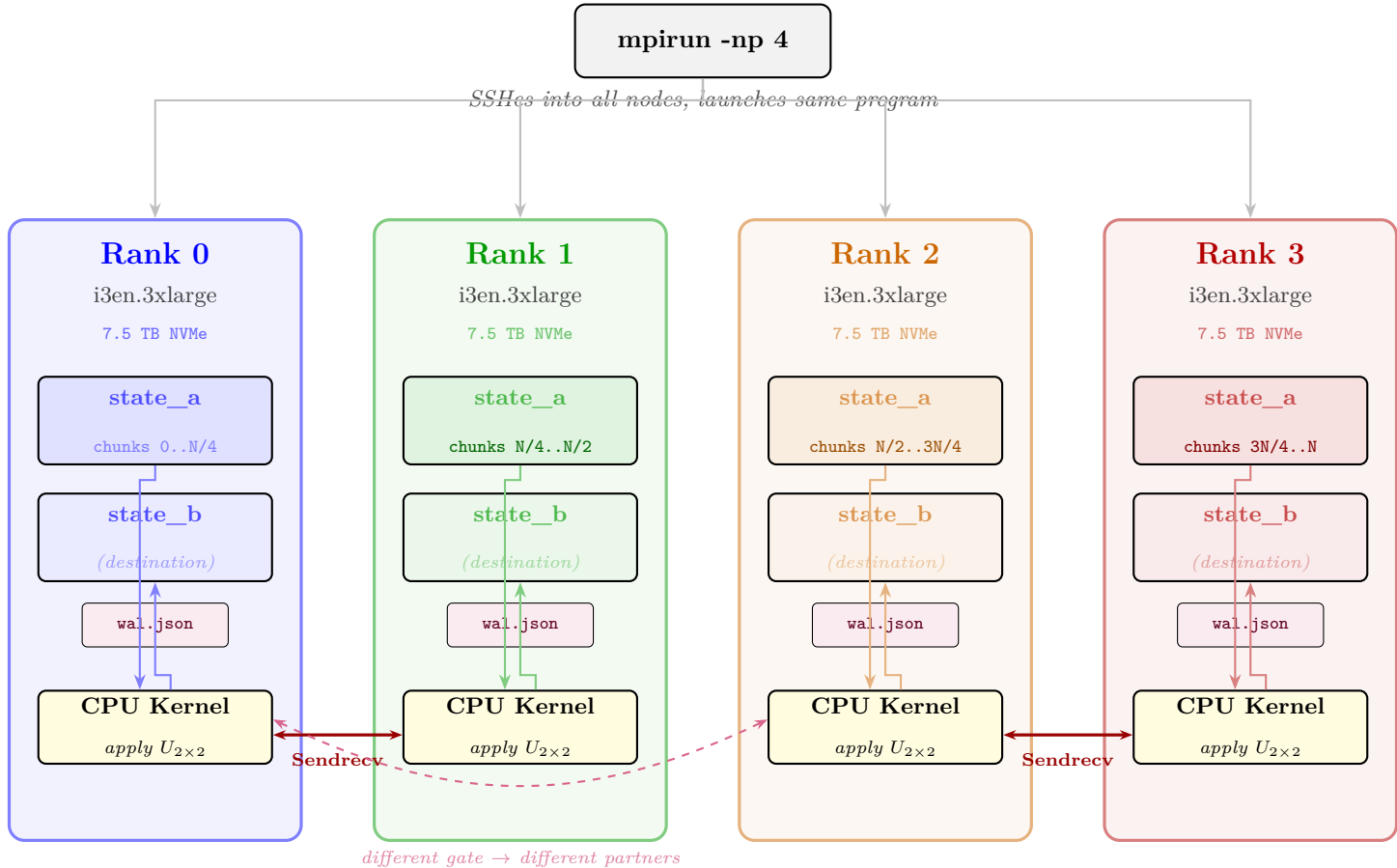


MPI Distributed Simulator Architecture



Step Protocol (per rank, in lockstep)

1. Wipe destination buffer (crash safety)
2. **Local gates**: read chunk from src → apply U → write to dst *zero network I/O*
3. **MPI gates**: Sendrecv chunk to partner rank → apply U across pair → write to dst *point-to-point*
4. **Barrier()** → update WAL → swap src/dst *at most 1 step lost on crash*